LANDAU-SIEGEL ZERO EXCLUSION FOR GL2 BY GL2 BY GL3 TRIPLE PRODUCTS OUTSIDE ADJOINT-TWIST EXCEPTIONS

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Allowed standard prerequisites (editorial).

- Automorphic tensor products, induction/base change and analytic Rankin-Selberg theory. Original complete bibliography and source-local auxiliary L-function arguments retained.

1. Introduction

1.1. **Background.** Let F be a number field. Let π be a cuspidal automorphic representation of $\mathrm{GL}_n(\mathbb{A}_F)$, $n \geq 1$. Denote by $L(s,\pi)$ the standard L-function associated to π. Let $c > 0$ be a positive number. We define a *Landau-Siegel zero* of $L(s,\pi)$, relative to c, to be a real zero β of $L(s,\pi)$ satisfying

$$1 - \frac{c}{\log C(\pi)} < \beta < 1.$$

Here $C(\pi)$ denotes the analytic conductor of $L(s,\pi)$.

Studying Landau-Siegel zeros of automorphic L-functions is a central topic in analytic number theory. For example, when $F = \mathbb{Q}$ and $\pi = \chi$ is a real primitive Dirichlet character, the possibility of existence of a Landau-Siegel zero of the Dirichlet L-function $L(s,\chi)$ is a long-standing open problem. Knowing the nonexistence or the existence of such a zero, on both sides, has deep consequences in other important arithmetic problems such as the distribution of prime numbers in arithmetic progressions [Dav00], the class number problem [Dav00], the twin prime conjecture [HB83], and the non-vanishing of L-functions on GL(2) [IS00], to name a few.

Although the problem of Landau-Siegel zeros of Dirichlet L-functions is extremely hard, eliminating Landau-Siegel zeros of higher degree automorphic L-functions is surprisingly more approachable. In the appendix of the ground-breaking paper [HL94], Goldfeld, Hoffstein and Lieman proved that if π is a non-dihedral cuspidal representation of $\mathrm{GL}_2(\mathbb{A}_{\mathbb{Q}})$, then $L(s, \mathrm{Ad}(\pi))$ has no Landau-Siegel zero. Here $\mathrm{Ad}(\pi)$ denotes the adjoint square lift of π, which is a self-dual cuspidal representation of $\mathrm{GL}_3(\mathbb{A}_{\mathbb{Q}})$ studied in [GJ78]. Inspired by their work, Landau-Siegel zeros of standard L-functions $L(s,\pi)$ were extensively studied later in several papers including [HR95][Ban97][Tho21][Luo23]. We will review some of these results in more details in Section 2.2.

Let π_1 and π_2 be cuspidal automorphic representations of $\mathrm{GL}_n(\mathbb{A}_F)$ and $\mathrm{GL}_m(\mathbb{A}_F)$, respectively. We similarly define a *Landau-Siegel zero* of the Rankin-Selberg L-function $L(s, \pi_1 \times \pi_2)$, relative to a positive constant $c > 0$, to be a real zero β of $L(s, \pi_1 \times \pi_2)$ satisfying

$$1 - \frac{c}{\log(C(\pi_1)C(\pi_2))} < \beta < 1.$$

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Landau-Siegel zeros of Rankin-Selberg L-functions were first studied by Ramakrishnan and Wang [RW03]. They proved that for cuspidal representations π_1 and π_2 of $\mathrm{GL}_2(\mathbb{A}_F)$, the Rankin-Selberg L-function $L(s, \pi_1 \times \pi_2)$ can only possibly have a Landau-Siegel zero in a few special cases. In particular, when the pair (π_1, π_2) is of *general type*, i.e., when π_1 and π_2 are non-dihedral and are not twist-equivalent, $L(s, \pi_1 \times \pi_2)$ has no Landau-Siegel zero. Landau-Siegel zeros of other types of Rankin-Selberg L-functions were studied in [Luo23][Tho25a][Tho25b]. We will also review some of these results in more details in Section 2.2.

1.2. **Main Results.** In this paper we study Landau-Siegel zeros of *triple product* L-functions on $\mathrm{GL}(2) \times \mathrm{GL}(2) \times \mathrm{GL}(2)$ and on $\mathrm{GL}(2) \times \mathrm{GL}(2) \times \mathrm{GL}(3)$.

Let π_1, π_2 and π_3 be cuspidal automorphic representations of $\mathrm{GL}_2(\mathbb{A}_F)$. The triple product L-function $L(s, \pi_1 \times \pi_2 \times \pi_3)$ was first studied in [Gar87] when $F = \mathbb{Q}$ and π_1, π_2, π_3 associate to holomorphic cusp forms. The construction was then generalized to any number field F in [PSR87]. Moreover, in view of the work [Ram00] on the modularity of Rankin-Selberg product on $\mathrm{GL}(2) \times \mathrm{GL}(2)$, the triple product L-function $L(s, \pi_1 \times \pi_2 \times \pi_3)$ can also be viewed as a Rankin-Selberg L-function on $\mathrm{GL}(4) \times \mathrm{GL}(2)$:

$$L(s, \pi_1 \times \pi_2 \times \pi_3) = L(s, (\pi_1 \boxtimes \pi_2) \times \pi_3),$$

whose analytic properties are well-known by [JPSS83].

We say the triple (π_1, π_2, π_3) is of

- *general type*, if π_1, π_2, π_3 are all non-dihedral, and π_i is not twist-equivalent to π_j for all $i \neq j$.
- *twist-equivalent type*, if π_1, π_2, π_3 are all non-dihedral, but π_i is twist-equivalent to π_j for some $i \neq j$.
- *dihedral type*, if at least one of π_1, π_2, π_3 is dihedral.

With these terminologies we are now ready to state our first main result.

Theorem A. *There exists an absolute effective constant $c > 0$, such that the triple product L-function $L(s, \pi_1 \times \pi_2 \times \pi_3)$ has no Landau-Siegel zero relative to c, except possibly in the following cases:*

(1) (π_1, π_2, π_3) is of dihedral type.
(2) (π_1, π_2, π_3) is of twist-equivalent type, and moreover π_1, π_2, π_3 are all twist-equivalent to each other.

Remark 1.1. *Theorem A is a simplified version of the full statement Theorem 3.1. See Theorem 3.1 for a precise description on the conditions when $L(s, \pi_1 \times \pi_2 \times \pi_3)$ could possibly have a Landau-Siegel zero. In particular, $L(s, \pi_1 \times \pi_2 \times \pi_3)$ has no Landau-Siegel zero if (π_1, π_2, π_3) is of general type.*

Remark 1.2. *To illustrate why there might be a Landau-Siegel zero when (π_1, π_2, π_3) is of dihedral type, let us assume π_1 is dihedral. Then $\pi_1 = I_K^F(\chi_1)$ is induced from a character χ_1 on some quadratic extension K/F. By the adjointness property of Rankin-Selberg L-functions we have*

$$L(s, \pi_1 \times \pi_2 \times \pi_3) = L(s, B_F^K(\pi_2) \times B_F^K(\pi_3) \otimes \chi_1),$$

where $B_F^K(\pi_2)$ and $B_F^K(\pi_3)$ are base changes from F to K of π_2 and π_3, respectively. Thus the triple product L-function reduces to a Rankin-Selberg L-function on $GL(2) \times GL(2)$ over

K, a case that has been studied in [RW03]. *According to* [RW03, Theorem A], *there are two reduced cases where there might be a Landau-Siegel zero.*

However, our argument in Section 3.3 gives more information on the possible Landau-Siegel zeros of $L(s, \pi_1 \times \pi_2 \times \pi_3)$ than those that can be obtained by directly applying [RW03, Theorem A]. *This is due to the special natures of triple product L-functions.*

Remark 1.3. *To see why there might be a Landau-Siegel zero when π_1, π_2, π_3 are all twist-equivalent, let us simply assume that $\pi_1 = \pi_2 = \pi_3 = \pi$ is a non-dihedral cuspidal representation of $GL_2(\mathbb{A}_F)$ with central character ω_π. In this case the triple product L-function decomposes as*

$$L(s, \pi \times \pi \times \pi) = L(s, Sym^3(\pi)) \cdot L(s, \pi \otimes \omega_\pi)^2.$$

However, eliminating Landau-Siegel zeros of the symmetric cube L-function $L(s, Sym^3(\pi))$ is a long-standing open problem, which is only known under the (very strong) hypothesis that certain twisted symmetric fifth power L-function is analytic in some interval $(t, 1)$, $t < 1$.

Let us proceed to describe our second main result. Let π_1, π_2 be cuspidal automorphic representations of $\mathrm{GL}_2(\mathbb{A}_F)$. Let π be a cuspidal automorphic representation of $\mathrm{GL}_3(\mathbb{A}_F)$. Then the triple product L-function $L(s, \pi_1 \times \pi_2 \times \pi)$ on $\mathrm{GL}(2) \times \mathrm{GL}(2) \times \mathrm{GL}(3)$ can likewise be viewed as a Rankin-Selberg L-function on $\mathrm{GL}(4) \times \mathrm{GL}(3)$:

$$L(s, \pi_1 \times \pi_2 \times \pi) = L(s, (\pi_1 \boxtimes \pi_2) \times \pi).$$

We say the triple (π_1, π_2, π) is of

- *general type*, if π_1, π_2 are non-dihedral, not twist-equivalent, and π is not twist-equivalent to either $\mathrm{Ad}(\pi_1)$ or $\mathrm{Ad}(\pi_2)$.
- *twist-equivalent type*, if π_1, π_2 are non-dihedral, but either (1) π_1 is twist-equivalent to π_2, or (2) π is twist-equivalent to $\mathrm{Ad}(\pi_1)$ or $\mathrm{Ad}(\pi_2)$.
- *dihedral type*, if at least one of π_1, π_2 is dihedral.

Our second main result is as follows.

Theorem B. *There exists an absolute effective constant $c > 0$ such that the triple product L-function $L(s, \pi_1 \times \pi_2 \times \pi)$ has no Landau-Siegel zero relative to c, except possibly in the case where π is twist-equivalent to $Ad(\pi_1)$ or $Ad(\pi_2)$.*

Remark 1.4. *See Theorem 4.1 for a more refined version of Theorem B, where we give a precise description on the conditions when $L(s, \pi_1 \times \pi_2 \times \pi)$ could possibly have a Landau-Siegel zero. In particular, $L(s, \pi_1 \times \pi_2 \times \pi)$ has no Landau-Siegel zero if (π_1, π_2, π) is of general type or dihedral type.*

Remark 1.5. *It is interesting to see that $L(s, \pi_1 \times \pi_2 \times \pi)$ has no Landau-Siegel zero if (π_1, π_2, π) is of dihedral type. To see why this is not obvious, let us assume $\pi_1 = I_K^F(\chi_1)$ is dihedral. Then we have*

$$L(s, \pi_1 \times \pi_2 \times \pi) = L(s, B_F^K(\pi_2) \times B_F^K(\pi) \otimes \chi_1).$$

Landau-Siegel zeros of the $GL(2) \times GL(3)$ Rankin-Selberg L-function on the right side have been studied in [Luo23]. *According to* [Luo23, Theorem 2], *there might be a Landau-Siegel zero if $B_F^K(\pi_2)$ is cuspidal non-dihedral, and $B_F^K(\pi)$ is twist-equivalent to $Ad(B_F^K(\pi_2))$. However, using a different argument we are able to show that there is no Landau-Siegel zero at all in this dihedral case. This argument, again, relies on special natures of triple product L-functions.*

Remark 1.6. *To see why there might be a Landau-Siegel zero when π is twist-equivalent to $Ad(\pi_1)$ or $Ad(\pi_2)$, let us assume that $\pi = Ad(\pi_1)$, and π_1, π_2 are not twist-equivalent. In this case we have*

$$L(s, \pi_1 \times \pi_2 \times \pi) = L(s, Sym^3(\pi_1) \times \pi_2 \otimes \omega_1^{-1}) \cdot L(s, \pi_1 \times \pi_2),$$

where ω_1 is the central character of π_1. Eliminating Landau-Siegel zeros of the Rankin-Selberg L-function $L(s, Sym^3(\pi_1) \times \pi_2 \otimes \omega_1^{-1})$ is an open problem.

In [Tho25b][Tho25a], Thorner addressed the importance of allowing an additional character twist to get a standard zero-free region. In our case, let π_1, π_2 be cuspidal automorphic representations on $\mathrm{GL}_2(\mathbb{A}_F)$ and π be a cuspidal automorphic representation on $\mathrm{GL}_2(\mathbb{A}_F)$ or $\mathrm{GL}_3(\mathbb{A}_F)$. Let η be an arbitrary (idele class) character on F. Then in view of the relation

$$L(s, \pi_1 \times \pi_2 \times \pi \otimes \eta) = L(s, \pi_1 \times \pi_2 \times (\pi \otimes \eta)),$$

we can study the Landau-Siegel zeros of $L(s, \pi_1 \times \pi_2 \times \pi \otimes \eta)$ by applying Theorem A and Theorem B to the triple $(\pi_1, \pi_2, \pi \otimes \eta)$. In particular, if (π_1, π_2, π) is of general type, then $(\pi_1, \pi_2, \pi \otimes \eta)$ is also of general type. By taking $\eta = |\cdot|^{it}$, $t \in \mathbb{R}$, we immediately conclude the following:

Theorem C. *Assume (π_1, π_2, π) is of general type. Then there exists an effective absolute constant $c > 0$, such that $L(s, \pi_1 \times \pi_2 \times \pi) \neq 0$ in the following region:*

$$\left\{ s = \sigma + it : \sigma > 1 - \frac{c}{\log(C(\pi_1)C(\pi_2)C(\pi)(|t|+3)^{[F:\mathbb{Q}]})} \right\}.$$

1.3. **Proof Strategy.** As is well-known to experts, the technical heart behind any attempt to eliminate Landau-Siegel zeros of an L-function $L(s, \pi)$ (or $L(s, \pi_1 \times \pi_2)$) is to construct an auxiliary L-function $L(s)$ with at most r Landau-Siegel zeros, in such a way that we can decompose $L(s)$ as

$$L(s) = L(s, \pi)^a \cdot L(s, \overline{\pi})^b \cdot L_{Res}(s),$$

with $a + b > r$, and the residual part $L_{Res}(s)$ is analytic in some interval $(t, 1)$, $t < 1$. If we have constructed such an $L(s)$, then any Landau-Siegel zero of $L(s, \pi)$ would be a Landau-Siegel zero of order at least $a + b$ of $L(s)$, which cannot exist because $L(s)$ has at most r Landau-Siegel zeros, and $r < a + b$.

For most of the work in this field, including the present paper, the auxiliary L-function is constructed as a special type of Rankin-Selberg L-function, $L(s) = L(s, \Pi \times \overline{\Pi})$, where Π is some carefully chosen isobaric automorphic representation. Thus if there is anything creative in this paper, it is the construction of Π, which, in the new context of triple product L-functions, exhibits some new features.

Recently, in his preprint [Tho25b] towards the study of Landau-Siegel zeros of Rankin-Selberg L-functions of symmetric power liftings of Hilbert modular forms over totally real fields, Thorner (for the first time) constructed auxiliary L-functions $L(s)$ that are not in the shape of $L(s, \Pi \times \overline{\Pi})$. In [Tho25a], he also studied Landau-Siegel zeros of Rankin-Selberg L-functions $L(s, \mathrm{Ad}(\pi_1) \times \mathrm{Ad}(\pi_2) \otimes \chi)$, where π_1, π_2 are cuspidal representations of $\mathrm{GL}_2(\mathbb{A}_F)$, and χ is a character on F. As we shall see later, our study of Landau-Siegel zeros of $\mathrm{GL}(2) \times \mathrm{GL}(2) \times \mathrm{GL}(3)$ triple product L-functions reduces to this type in some particular cases. So we would like to express our gratefulness to Prof. Jesse Thorner for his great work.

This paper is organized as follows. In Section 2 we review some preliminaries, including some instances of Langlands functoriality and some known cases of the nonexistence of

Landau-Siegel zeros that will be used throughout this paper. In Section 3 and Section 4 we prove Theorem A and Theorem B, respectively.

2. Preliminaries

2.1. **Langlands Functoriality.** In this section we review some known instances of Langlands functoriality that are crucial to our work. First let us introduce some notations. We use $\mathcal{A}_0(n,F)$ (resp. $\mathcal{A}(n,F)$) to denote the set of cuspidal (resp. isobaric) automorphic representations of $\mathrm{GL}_n(\mathbb{A}_F)$ with unitary central characters ω_π. For $\pi\in\mathcal{A}(n,F)$ we denote by $\overline{\pi}$ the *dual* representation of π. We say π is *self-dual*, if $\pi\cong\overline{\pi}$. We say π is *monomial*, if $\pi\cong\pi\otimes\eta$ for some non-trivial character η on F. We say $\pi_1,\pi_2\in\mathcal{A}_0(n,F)$ are *twist-equivalent*, if $\pi_1\cong\pi_2\otimes\mu$ for some character μ on F.

Let us start with base change and automorphic induction. Let K/F be a cyclic extension of number fields of prime degree ℓ. Let $n\geq 1$ be a positive integer. The base change map

$$B_F^K:\mathcal{A}(n,F)\to\mathcal{A}(n,K),\quad \pi\mapsto B_F^K(\pi)$$

and the automorphic induction map

$$I_K^F:\mathcal{A}(n,K)\to\mathcal{A}(n\ell,F),\quad \alpha\mapsto I_K^F(\alpha)$$

were constructed in [AC89]. These maps satisfy the following properties:

Lemma 2.1. *Let $\theta=\theta_{K/F}$ be a generator of the Galois group $Gal(K/F)$. Let $\eta=\eta_{K/F}$ be the character on F associated to K/F. Then*

(1) Let $\pi\in\mathcal{A}(n\ell,F)$. Then π lies in the image of I_K^F if and only if $\pi\cong\pi\otimes\eta$.
(2) Let $\alpha\in\mathcal{A}(n,K)$. Then α lies in the image of B_F^K if and only if $\alpha\cong\alpha\circ\theta$.
(3) For $\pi\in\mathcal{A}(n,F)$ and $\alpha\in\mathcal{A}(m,K)$, we have

$$L(s,\pi\times I_K^F(\alpha))=L(s,B_F^K(\pi)\times\alpha).$$

(4) Let $\pi\in\mathcal{A}_0(n,F)$ be cuspidal. Then $B_F^K(\pi)$ is cuspidal if and only if π is not isomorphic to $\pi\otimes\eta$. Moreover, we have

$$I_K^F(B_F^K(\pi))\cong\boxplus_{j=0}^{\ell-1}\pi\otimes\eta^j.$$

(5) Let $\alpha\in\mathcal{A}_0(n,K)$ be cuspidal. Then $I_K^F(\alpha)$ is cuspidal if and only if α is not isomorphic to $\alpha\circ\theta$. Moreover, we have

$$B_F^K(I_K^F(\alpha))\cong\boxplus_{j=0}^{\ell-1}\alpha\circ\theta^j.$$

We then review functoriality results of symmetric power liftings. Let $\pi\in\mathcal{A}_0(2,F)$. Denote by $\mathrm{Sym}^m(\pi)$ the symmetric m^{th}-power lifting of π. We set $\mathrm{Ad}(\pi):=\mathrm{Sym}^2(\pi)\otimes\omega_\pi^{-1}$, $A^3(\pi):=\mathrm{Sym}^3(\pi)\otimes\omega_\pi^{-1}$ and $A^4(\pi):=\mathrm{Sym}^4(\pi)\otimes\omega_\pi^{-2}$. We say π is

- *dihedral*, if π is monomial, or equivalently, if $\pi=I_K^F(\chi)$ is an automorphic induction of a character χ on some quadratic extension K/F.
- *tetrahedral*, if $\mathrm{Sym}^2(\pi)$ is cuspidal and monomial.
- *octahedral*, if $\mathrm{Sym}^3(\pi)$ is cuspidal and monomial.
- of *solvable polyhedral type*, if π is either dihedral or tetrahedral or octahedral.

Below is a summary of [GJ78, Theorem 9.3],[KS02b, Theorem B],[KS02a, Theorem 2.2.2],[Kim03, Theorem B],[KS02a, Theorem 3.3.7] and [NT25, Theorem A]:

Lemma 2.2. *Let $\pi\in\mathcal{A}_0(2,F)$. Then we have*

(1) *$Sym^2(\pi) \in \mathcal{A}(3, F)$. $Sym^2(\pi)$ is not cuspidal if and only if π is dihedral. Moreover, $Ad(\pi)$ is self-dual.*
(2) *$Sym^3(\pi) \in \mathcal{A}(4, F)$. $Sym^3(\pi)$ is not cuspidal if and only if π is of dihedral or tetrahedral type. Moreover, when π is of tetrahedral type, write $Ad(\pi) \cong Ad(\pi) \otimes \eta$ for some cubic character η on F. In this case we have*
$$A^3(\pi) = (\pi \otimes \eta) \boxplus (\pi \otimes \eta^2).$$
(3) *$Sym^4(\pi) \in \mathcal{A}(5, F)$. $Sym^4(\pi)$ is not cuspidal if and only if π is of solvable polyhedral type. Moreover,*
   (a) *When π is of tetrahedral type, write $Ad(\pi) \cong Ad(\pi) \otimes \eta$ for some cubic character η on F. In this case we have*
$$A^4(\pi) = Ad(\pi) \boxplus \eta \boxplus \eta^2.$$
   (b) *When π is of octahedral type, write $Sym^3(\pi) \cong Sym^3(\pi) \otimes \eta$ for some quadratic character η on F. Let K/F be the quadratic extension associated with η. Then $Ad(B_F^K(\pi)) \cong Ad(B_F^K(\pi)) \otimes \chi$ for some cubic character χ on K. In this case we have*
$$A^4(\pi) = I_K^F(\chi^{-1}) \boxplus (Ad(\pi) \otimes \eta).$$
(4) *Assume F is totally real, and π is non-dihedral regular algebraic. Then $Sym^{m-1}(\pi) \in \mathcal{A}_0(m, F)$ for all $m \geq 2$.*

Finally we review functoriality results of Rankin-Selberg products on $\mathrm{GL}(2) \times \mathrm{GL}(2)$ and on $\mathrm{GL}(2) \times \mathrm{GL}(3)$. Below is a summary of [Ram00, Theorem M], [KS02b, Theorem A] and [RW04, Theorem 3.1]:

Lemma 2.3. *Let $\pi_1, \pi_2 \in \mathcal{A}_0(2, F)$. Let $\pi \in \mathcal{A}_0(3, F)$. Then we have*
(1) *$\pi_1 \boxtimes \pi_2 \in \mathcal{A}(4, F)$. Moreover, $\pi_1 \boxtimes \pi_2$ is cuspidal if and only if one the following happens:*
   (a) *π_1, π_2 are both non-dihedral, and π_1 is not twist-equivalent to π_2.*
   (b) *One of π_1, π_2, say π_1, is dihedral. Then $\pi_1 = I_K^F(\chi_1)$ for a character χ_1 on some quadratic extension K/F. And the base change $B_F^K(\pi_2)$ is cuspidal and not isomorphic to $B_F^K(\pi_2) \otimes (\chi_1 \circ \theta_{K/F})\chi_1^{-1}$.*
(2) *$\pi_1 \boxtimes \pi \in \mathcal{A}(6, F)$. Moreover, $\pi_1 \boxtimes \pi$ is not cuspidal if and only if one of the following happens:*
   (a) *π_1 is non-dihedral, and π is twist-equivalent to $Ad(\pi_1)$.*
   (b) *π_1 is dihedral, $L(s, \pi) = L(s, \chi)$ for a character χ on a cubic, non-normal extension K/F, and the base change $B_F^K(\pi)$ satisfies $L(s, B_F^K(\pi)) = L(s, \psi)$ for a character ψ on a quadratic extension L/K.*

At the end of this section we prepare the following lemma for future reference:

Lemma 2.4. *Let $\pi_1, \pi_2 \in \mathcal{A}_0(2, F)$ be non-dihedral. Suppose that $Ad(\pi_1)$ and $Ad(\pi_2)$ are twist-equivalent. Then π_1 and π_2 are also twist-equivalent.*

Proof. Suppose $\mathrm{Ad}(\pi_1) \cong \mathrm{Ad}(\pi_2) \otimes \eta$ for some character η on F. Computing the central characters of both sides we have
$$1 = \omega_{\mathrm{Ad}(\pi_1)} = \omega_{\mathrm{Ad}(\pi_2)\otimes\eta} = \omega_{\mathrm{Ad}(\pi_2)}\eta^3 = \eta^3.$$
If $\eta = 1$, then $\mathrm{Ad}(\pi_1) \cong \mathrm{Ad}(\pi_2)$, which would imply π_1 and π_2 are twist-equivalent by the multiplicity one theorem [Ram00, Theorem 4.1.2]. If $\eta \neq 1$, then for each finite place v where

π_1, π_2 and η are unramified we have the following two sets of local parameters of $\mathrm{Ad}(\pi_1)$ and $\mathrm{Ad}(\pi_2)\otimes\eta$ at v

$$\{\alpha_{1,v}, 1, \alpha_{1,v}^{-1}\} = \{\alpha_{2,v}\eta_v, \eta_v, \alpha_{2,v}^{-1}\eta_v\}$$

are the same. Thus either $\eta_v = \alpha_{1,v}$ or $\eta_v = \alpha_{1,v}^{-1}$. By symmetry we may assume that $\eta_v = \alpha_{1,v}$. Thus we have

$$\{\alpha_{1,v}\eta_v, \eta_v, \alpha_{1,v}^{-1}\eta_v\} = \{\alpha_{1,v}, 1, \alpha_{1,v}^2\} = \{\alpha_{1,v}, 1, \alpha_{1,v}^{-1}\},$$

which implies that

$$\mathrm{Ad}(\pi_{1,v}) \cong \mathrm{Ad}(\pi_{1,v})\otimes\eta_v \cong \mathrm{Ad}(\pi_{2,v})\otimes\eta_v.$$

Thus we have

$$\mathrm{Ad}(\pi_{1,v}) \cong \mathrm{Ad}(\pi_{2,v})$$

for all but finitely many v. By [Ram00, Theorem 4.1.2] this would also imply that π_1 and π_2 are twist equivalent. □

2.2. **Known Cases of Landau-Siegel Zeros.** In this section we review some known cases where Landau-Siegel zeros can be eliminated. Our study of Landau-Siegel zeros of triple product L-functions reduces to them in some special cases.

Let us first state results on standard L-functions. Below is a summary of [HR95, Corollary 3.2], [Luo23, Theorem 3], [HR95, Theorem C] and [Ban97, Theorem 1]:

Lemma 2.5. *Let $\pi \in \mathcal{A}_0(n, F)$. Then $L(s,\pi)$ has no Landau-Siegel zero in the following cases:*

(1) π is non-self-dual.
(2) $n \geq 2$ and π is monomial.
(3) $n = 2, 3$.

We then state results on Rankin-Selberg L-functions. Below is a summary of [RW03, Theorem A],[Luo23, Theorem 1] and [Luo23, Theorem 2]:

Lemma 2.6. *Let $\pi_1, \pi_2 \in \mathcal{A}_0(2, F)$. Let $\pi \in \mathcal{A}_0(3, F)$. Then we have*

(1) $L(s, \pi_1 \times \pi_2)$ has no Landau-Siegel zero except possibly in the following cases:
  (a) π_1, π_2 are non-dihedral, and π_1 is twist-equivalent to π_2.
  (b) π_1 and π_2 are both dihedral, and $\pi_1 = I_K^F(\chi_1), \pi_2 = I_K^F(\chi_2)$ for characters χ_1, χ_2 on a single quadratic extension K/F.
(2) $L(s, \pi_1 \times \pi)$ has no Landau-Siegel zero except possibly in the case when π_1 is non-dihedral, and π is twist-equivalent to $Ad(\pi_1)$.

All the results stated above rely on the following lemma, which is a combination of [HR95, Lemma a] and [HL94, Appendix, Lemma].

Lemma 2.7. *Let $\Pi \in \mathcal{A}(n, F)$. Write $\Pi = \boxplus_{j=1}^J e_j\pi_j$ as an isobaric sum of cuspidal representations, where π_i is not isomorphic to π_j when $i \neq j$, and $e_j \geq 1$. Then there exists an effective constant $c = c(n, e_j)$ depending only on n and the e_j's, such that $L(s, \Pi \times \overline{\Pi})$ has at most $e = \sum_{j=1}^J e_j^2$ many Landau-Siegel zeros relative to c. Moreover, if $L(s, \Pi \times \overline{\Pi})$ decomposes as*

$$L(s, \Pi \times \overline{\Pi}) = L_1(s)^a \cdot L_2(s)^b \cdot L_{Res}(s),$$

where $a + b > e$, $L_1(s)$ shares the same real zeros with $L_2(s)$, and $L_{Res}(s)$ is analytic in some interval $(t, 1)$, $t < 1$, then $L_1(s)$ has no Landau-Siegel zero.

At the end of this section let us prove the following simple lemma that will be used later:

Lemma 2.8. *Let $\pi_1 \in \mathcal{A}_0(n,F)$ and $\pi_2 \in \mathcal{A}_0(m,F)$. Then $L(s,\pi_1\times\pi_2)$ has no Landau-Siegel zero if one of the following happens:*

(1) π_1 is self-dual, and π_2 is non-self-dual.
(2) π_1 is monomial with $\pi_1 \cong \pi_1\otimes\eta$ for a non-trivial character η on F, and π_2 is not isomorphic to $\pi_2\otimes\eta$.

Proof. (1) Since π_1 is self-dual and π_2 is not, no two of the three representations $\pi_1,\pi_2,\overline{\pi_2}$ are isomorphic. Set $\Pi=\pi_1\boxplus\pi_2\boxplus\overline{\pi_2}$. Note that $L(s,\Pi\times\overline{\Pi})$ decomposes as

$$L(s,\Pi\times\overline{\Pi}) = L(s,\pi_1\times\pi_2)^2\cdot L(s,\pi_1\times\overline{\pi_2})^2 \cdot L(s,\pi_1\times\pi_1)\cdot L(s,\pi_2\times\pi_2)\cdot L(s,\overline{\pi_2}\times\overline{\pi_2})\cdot L(s,\pi_2\times\overline{\pi_2})^2.$$

Since π_1 is self-dual, $L(s,\pi_1\times\pi_2)$ and $L(s,\pi_1\times\overline{\pi_2})$ have the same real zeros. Thus $L(s,\pi_1\times\pi_2)$ has no Landau-Siegel zero by Lemma 2.7.

(2) Since $\pi_1\cong\pi_1\otimes\eta$ and π_2 is not isomorphic to $\pi_2\otimes\eta$, no two of $\overline{\pi_1},\pi_2,\pi_2\otimes\eta$ are isomorphic. Set $\Pi=\overline{\pi_1}\boxplus\pi_2\boxplus(\pi_2\otimes\eta)$. In view of the relation $\pi_1\cong\pi_1\otimes\eta$, $L(s,\Pi\times\overline{\Pi})$ decomposes as

$$L(s,\Pi\times\overline{\Pi}) = L(s,\pi_1\times\pi_2)^2\cdot L(s,\overline{\pi_1}\times\overline{\pi_2})^2 \cdot L(s,\pi_1\times\overline{\pi_1})\cdot L(s,\pi_2\times\overline{\pi_2})^2\cdot L(\pi_2\times\overline{\pi_2}\otimes\eta)\cdot L(s,\pi_2\times\overline{\pi_2}\otimes\eta^{-1}).$$

Thus $L(s,\pi_1\times\pi_2)$ has no Landau-Siegel zero by Lemma 2.7.

□

3. Proof of Theorem A

The goal of this section is to prove the following refined version of Theorem A:

Theorem 3.1. *Let $\pi_1,\pi_2,\pi_3\in\mathcal{A}_0(2,F)$ with central characters $\omega_1,\omega_2,\omega_3$ respectively. Then $L(s,\pi_1\times\pi_2\times\pi_3)$ has no Landau-Siegel zero except possibly in the following cases:*

(1) There exists a quadratic extension K/F such that $\pi_i = I_K^F(\chi_i)$ for some character χ_i on K, for each $i=1,2,3$. In this case there are at most two Landau-Siegel zeros coming from

$$L(s,\chi_1\chi_2\chi_3)L(s,\chi_1\chi_2^\theta\chi_3)L(s,\chi_1\chi_2\chi_3^\theta)L(s,\chi_1\chi_2^\theta\chi_3^\theta).$$

Here $\theta=\theta_{K/F}$ is the non-trivial element in $Gal(K/F)$, and wet set $\chi_i^\theta := \chi_i\circ\theta$. Moreover, there is at most one Landau-Siegel zero if $(\chi_2\chi_3)^\theta\neq\chi_2\chi_3$.

(2) There exists a quadratic extension K/F such that one of π_1,π_2,π_3, say π_1, is induced from some character χ_1 on K, and $B_F^K(\pi_2)\otimes\chi_1$ and $B_F^K(\pi_3)$ are both cuspidal and dihedral, induced form characters ξ_2,ξ_3 on L respectively, where L/K is the quadratic extension associated with the quadratic character $\chi_1^\theta\chi_1^{-1}$ on K. In this case there is at most one Landau-Siegel zero coming from

$$L(s,\xi_3\xi_2)L(s,\xi_3\xi_2^{\theta_{L/K}}).$$

(3) π_1,π_2,π_3 are not of solvable polyhedral type, and are all twist-equivalent. Moreover, suppose $\pi_2\cong\pi_1\otimes\eta$ and $\pi_3\cong\pi_1\otimes\mu$ for characters η and μ on F. Then there is no Landau-Siegel zero if the twisted symmetric fifth power L-function

$$L(s,\pi_1;Sym^5\otimes\omega_1^{-1}\eta\mu)$$

has no pole in some interval $(t, 1)$, $t < 1$.

Remark 3.1. *The condition in Theorem 3.1 (3) is satisfied if* F *is totally real, and* π_1 *is regular algebraic by Lemma 2.2 (4). Thus there is no Landau-Siegel zero in this case.*

3.1. **General Type.** Assume (π_1, π_2, π_3) is of general type. Recall that this means π_1, π_2, π_3 are all non-dihedral, and π_i is not twist-equivalent to π_j for all $i \neq j$. We set

$$\Pi := (\pi_1 \boxtimes \pi_2) \boxplus \overline{\pi_3} \boxplus (\mathrm{Ad}(\pi_1) \boxtimes \overline{\pi_3}).$$

Now since π_1, π_2 are non-dihedral and are not twist-equivalent, $\pi_1 \boxtimes \pi_2$ is cuspidal by Lemma 2.3 (1). We then claim that $\mathrm{Ad}(\pi_1) \boxtimes \overline{\pi_3}$ is also cuspidal. Suppose not, then by Lemma 2.3 (2), $\mathrm{Ad}(\pi_1)$ is twist-equivalent to $\mathrm{Ad}(\overline{\pi_3}) = \mathrm{Ad}(\pi_3)$. By Lemma 2.4 this would imply that π_1, π_2 are twist-equivalent, a contradiction!

Now the three representations $\pi_1 \boxtimes \pi_2, \overline{\pi_3}$ and $\mathrm{Ad}(\pi_1) \boxtimes \overline{\pi_3}$ are pairwise non-isomorphic, since they have different degrees. We have

$$\begin{aligned} L(s, \Pi \times \overline{\Pi}) &= L(s, \pi_1 \times \pi_2 \times \pi_3) \cdot L(s, \overline{\pi_1} \times \overline{\pi_2} \times \overline{\pi_3}) \\ &\quad \cdot L(s, (\pi_1 \boxtimes \pi_2) \times (\mathrm{Ad}(\pi_1) \boxtimes \pi_3)) \cdot L(s, (\overline{\pi_1} \boxtimes \overline{\pi_2}) \times (\mathrm{Ad}(\pi_1) \boxtimes \overline{\pi_3})) \\ &\quad \cdot L(s, \overline{\pi_3} \times (\mathrm{Ad}(\pi_1) \boxtimes \pi_3))^2 \\ &\quad \cdot L(s, (\pi_1 \boxtimes \pi_2) \times (\overline{\pi_1} \boxtimes \overline{\pi_2})) \cdot L(s, \pi_3 \times \overline{\pi_3}) \cdot L(s, (\mathrm{Ad}(\pi_1) \boxtimes \pi_3) \times (\mathrm{Ad}(\pi_1) \boxtimes \overline{\pi_3})). \end{aligned}$$

Note that we can further decompose $L(s, (\pi_1 \boxtimes \pi_2) \times (\mathrm{Ad}(\pi_1) \boxtimes \pi_3))$ as

$$\begin{aligned} L(s, (\pi_1 \boxtimes \pi_2) \times (\mathrm{Ad}(\pi_1) \boxtimes \pi_3)) &= L(s, (\pi_1 \boxtimes \mathrm{Ad}(\pi_1)) \times (\pi_2 \boxtimes \pi_3)) \\ &= L(s, (\pi_1 \boxplus A^3(\pi_1)) \times (\pi_2 \boxtimes \pi_3)) \\ &= L(s, \pi_1 \times \pi_2 \times \pi_3) \cdot L(s, A^3(\pi_1) \times (\pi_2 \boxtimes \pi_3)). \end{aligned}$$

Thus $L(s, \Pi \times \overline{\Pi})$ is equal to

$$\begin{aligned} L(s, \Pi \times \overline{\Pi}) &= L(s, \pi_1 \times \pi_2 \times \pi_3)^2 \cdot L(s, \overline{\pi_1} \times \overline{\pi_2} \times \overline{\pi_3})^2 \\ &\quad \cdot L(s, A^3(\pi_1) \times (\pi_2 \boxtimes \pi_3)) \cdot L(s, \overline{A^3(\pi_1)} \times (\overline{\pi_2} \boxtimes \overline{\pi_3})) \\ &\quad \cdot L(s, \overline{\pi_3} \times (\mathrm{Ad}(\pi_1) \boxtimes \pi_3))^2 \\ &\quad \cdot L(s, (\pi_1 \boxtimes \pi_2) \times (\overline{\pi_1} \boxtimes \overline{\pi_2})) \cdot L(s, \pi_3 \times \overline{\pi_3}) \cdot L(s, (\mathrm{Ad}(\pi_1) \boxtimes \pi_3) \times (\mathrm{Ad}(\pi_1) \boxtimes \overline{\pi_3})). \end{aligned}$$

By Lemma 2.7, $L(s, \pi_1 \times \pi_2 \times \pi_3)$ has no Landau-Siegel zero.

3.2. **Twist-equivalent Type.** Assume (π_1, π_2, π_3) is of twist-equivalent type. Recall that this means π_1, π_2, π_3 are all non-dihedral, but π_i is twist-equivalent to π_j for some $i \neq j$. Without loss of generality let us assume that π_1 and π_2 are twist-equivalent. Write $\pi_2 \cong \pi_1 \otimes \eta$ for some character η on F. Then $L(s, \pi_1 \times \pi_2 \times \pi_3)$ decomposes as

$$\begin{aligned} L(s, \pi_1 \times \pi_2 \times \pi_3) &= L(s, \pi_1 \times (\pi_1 \otimes \eta) \times \pi_3) \\ &= L(s, (\pi_1 \boxtimes \pi_1) \times \pi_3 \otimes \eta) \\ &= L(s, (\mathrm{Ad}(\pi_1) \otimes \omega_1 \boxplus \omega_1) \times \pi_3 \otimes \eta) \\ &= L(s, \mathrm{Ad}(\pi_1) \times \pi_3 \otimes \omega_1\eta) \cdot L(s, \pi_3 \otimes \omega_1\eta). \end{aligned}$$

Now $L(s, \pi_3 \otimes \omega_1\eta)$ has no Landau-Siegel zero by Lemma 2.5 (3). Moreover, by Lemma 2.6 (2), $L(s, \mathrm{Ad}(\pi_1) \times \pi_3 \otimes \omega_1\eta)$ can only possibly have Landau-Siegel zeros when $\mathrm{Ad}(\pi_1)$ is twist-equivalent to $\mathrm{Ad}(\pi_3)$, which would imply that π_1 is twist-equivalent to π_3 by Lemma

2.4. As a conclusion, $L(s,\pi_1\times\pi_2\times\pi_3)$ can only possibly have Landau-Siegel zeros if π_1,π_2,π_3 are all twist-equivalent to each other.

From now on we may further assume $\pi_3\cong\pi_1\otimes\mu$ for some character μ on F, and study Landau-Siegel zeros of $L(s,\mathrm{Ad}(\pi_1)\times\pi_3\otimes\omega_1\eta)$. In this case we have

$$\begin{aligned}L(s,\mathrm{Ad}(\pi_1)\times\pi_3\otimes\omega_1\eta)&=L(s,\mathrm{Ad}(\pi_1)\times(\pi_1\otimes\mu)\otimes\omega_1\eta)\\&=L(s,(\mathrm{Ad}(\pi_1)\boxtimes\pi_1)\otimes\omega_1\eta\mu)\\&=L(s,(\pi_1\boxplus A^3(\pi_1))\otimes\omega_1\eta\mu)\\&=L(s,\pi_1\otimes\omega_1\eta\mu)\cdot L(s,A^3(\pi_1)\otimes\omega_1\eta\mu).\end{aligned}$$

Now $L(s,\pi_1\otimes\omega_1\eta\mu)$ has no Landau-Siegel zero by Lemma 2.5 (3). To further discuss the Landau-Siegel zeros of $L(s,A^3(\pi_1)\otimes\omega_1\eta\mu)$ we consider the following cases:

(1) Assume π_1 is tetrahedral, i.e., $\mathrm{Ad}(\pi_1)$ is cuspidal and monomial. Write $\mathrm{Ad}(\pi_1)\cong\mathrm{Ad}(\pi_1)\otimes\nu$ for some cubic character ν on F. Then by Lemma 2.2 (2), $A^3(\pi_1)$ is not cuspidal and decomposes as

$$A^3(\pi_1)=(\pi_1\otimes\nu)\boxplus(\pi_1\otimes\nu^2).$$

Thus we have

$$L(s,A^3(\pi_1)\otimes\omega_1\eta\mu)=L(s,\pi_1\otimes\omega_1\eta\mu\nu)\cdot L(s,\pi_1\otimes\omega_1\eta\mu\nu^2),$$

which has no Landau-Siegel zero by Lemma 2.5 (3).

(2) Assume π_1 is octahedral, i.e., $\mathrm{Sym}^3(\pi_1)$ is cuspidal and monomial. Thus $A^3(\pi_1)\otimes\omega_1\eta\mu=\mathrm{Sym}^3(\pi_1)\otimes\eta\mu$ is also cuspidal and monomial. Therefore $L(s,A^3(\pi_1)\otimes\omega_1\eta\mu)$ has no Landau-Siegel zero by Lemma 2.5 (2).

(3) Assume π_1 is not of solvable polyhedral type. Then $\mathrm{Sym}^3(\pi_1)$ is cuspidal, and so is $\mathrm{Sym}^3(\pi_1)\otimes\eta\mu$. We construct

$$\Pi:=1\boxplus\mathrm{Ad}(\pi_1)\boxplus\mathrm{Sym}^3(\pi_1)\otimes\eta\mu.$$

Then we have

$$\begin{aligned}L(s,\Pi\times\overline{\Pi})=&L(s,\mathrm{Sym}^3(\pi_1)\otimes\eta\mu)\cdot L(s,\overline{\mathrm{Sym}^3(\pi_1)\otimes\eta\mu})\\&\cdot L(s,\mathrm{Ad}(\pi_1)\times(\mathrm{Sym}^3(\pi_1)\otimes\eta\mu))\cdot L(s,\mathrm{Ad}(\pi_1)\times(\overline{\mathrm{Sym}^3(\pi_1)\otimes\eta\mu}))\\&\cdot L(s,\mathrm{Ad}(\pi_1))^2\cdot\zeta_F(s)\cdot L(s,\mathrm{Ad}(\pi_1)\times\mathrm{Ad}(\pi_1))\\&\cdot L(s,(\mathrm{Sym}^3(\pi_1)\otimes\eta\mu)\times(\overline{\mathrm{Sym}^3(\pi_1)\otimes\eta\mu})).\end{aligned}$$

Note that $L(s,\mathrm{Ad}(\pi_1)\times(\mathrm{Sym}^3(\pi_1)\otimes\eta\mu))$ decomposes as

$$\begin{aligned}L(s,\mathrm{Ad}(\pi_1)\times(\mathrm{Sym}^3(\pi_1)\otimes\eta\mu))=&L(s,\mathrm{Sym}^3(\pi_1)\otimes\eta\mu)\\&\cdot L(s,\pi_1\otimes\omega_1\eta\mu)\cdot L(s,\pi_1;\mathrm{Sym}^5\otimes\omega_1^{-1}\eta\mu).\end{aligned}$$

Thus $L(s, \Pi \times \overline{\Pi})$ equals

$$\begin{aligned} L(s, \Pi \times \overline{\Pi}) = & L(s, \mathrm{Sym}^3(\pi_1) \otimes \eta\mu)^2 \cdot L(s, \overline{\mathrm{Sym}^3(\pi_1) \otimes \eta\mu})^2 \\ & \cdot L(s, \pi_1 \otimes \omega_1 \eta\mu) \cdot L(s, \overline{\pi_1 \otimes \omega_1 \eta\mu}) \\ & \cdot L(s, \pi_1; \mathrm{Sym}^5 \otimes \omega_1^{-1}\eta\mu) \cdot L(s, \overline{\pi_1}; \mathrm{Sym}^5 \otimes \omega_1 \eta^{-1}\mu^{-1}) \\ & \cdot L(s, \mathrm{Ad}(\pi_1))^2 \cdot \zeta_F(s) \cdot L(s, \mathrm{Ad}(\pi_1) \times \mathrm{Ad}(\pi_1)) \\ & \cdot L(s, (\mathrm{Sym}^3(\pi_1) \otimes \eta\mu) \times (\overline{\mathrm{Sym}^3(\pi_1) \otimes \eta\mu})). \end{aligned}$$

By Lemma 2.7, $L(s, \mathrm{Sym}^3(\pi_1) \otimes \eta\mu)$ has no Landau-Siegel zero if $L(s, \pi_1; \mathrm{Sym}^5 \otimes \omega_1^{-1}\eta\mu)$ is analytic in some interval $(t, 1)$, $t < 1$. This proves Theorem 3.1 (3).

3.3. **Dihedral Type.** Assume (π_1, π_2, π_3) is of dihedral type. Recall this means at least one of π_1, π_2, π_3, say π_1, is dihedral. Write $\pi_1 = I_K^F(\chi_1)$ for a character χ_1 on some quadratic extension K/F. Then we have

$$\pi_1 \boxtimes \pi_2 = I_K^F(\chi_1) \boxtimes \pi_2 = I_K^F(B_F^K(\pi_2) \otimes \chi_1).$$

By Lemma 2.1 we have

$$\begin{aligned} L(s, \pi_1 \times \pi_2 \times \pi_3) &= L(s, I_K^F(B_F^K(\pi_2) \otimes \chi_1) \times \pi_3) \\ &= L(s, B_F^K(\pi_2) \times B_F^K(\pi_3) \otimes \chi_1). \end{aligned}$$

We discuss Landau-Siegel zeros of the $\mathrm{GL}(2)\times\mathrm{GL}(2)$ Rankin-Selberg L-function $L(s, B_F^K(\pi_2)\times B_F^K(\pi_3) \otimes \chi_1)$ by the cuspidality of $B_F^K(\pi_2)$ and $B_F^K(\pi_3)$.

(1) Assume exactly one of $B_F^K(\pi_2)$ and $B_F^K(\pi_3)$ is cuspidal, say $B_F^K(\pi_2)$ is cuspidal, but $B_F^K(\pi_3)$ is not. By Lemma 2.1 we know that $\pi_3 = I_K^F(\chi_3)$ for some character χ_3 on K. In this case we have

$$\begin{aligned} L(s, \pi_1 \times \pi_2 \times \pi_3) &= L(s, B_F^K(\pi_2) \times B_F^K(\pi_3) \otimes \chi_1) \\ &= L(s, (B_F^K(\pi_2) \otimes \chi_1) \times (\chi_3 \boxplus \chi_3^\theta)) \\ &= L(s, B_F^K(\pi_2) \otimes \chi_1\chi_3) \cdot L(s, B_F^K(\pi_2) \otimes \chi_1\chi_3^\theta), \end{aligned}$$

which has no Landau-Siegel zero by Lemma 2.5 (3).

(2) Assume that both $B_F^K(\pi_2)$ and $B_F^K(\pi_3)$ are cuspidal. Let $\eta = \eta_{K/F}$ be the quadratic character on F associated with K/F. Then by Lemma 2.1 we know that π_2 is not isomorphic to $\pi_2 \otimes \eta$, and π_3 is not isomorphic to $\pi_3 \otimes \eta$.

(a) Assume further that at least one of $\pi_1 \boxtimes \pi_3$ and $\pi_1 \boxtimes \pi_2$, say $\pi_1 \boxtimes \pi_3$, is cuspidal. Then we have

$$(\pi_1 \boxtimes \pi_3) \otimes \eta = (\pi_1 \otimes \eta) \boxtimes \pi_3 = \pi_1 \boxtimes \pi_3.$$

Since π_2 is not isomorphic to $\pi_2 \otimes \eta$, the Rankin-Selberg L-function

$$L(s, (\pi_1 \boxtimes \pi_3) \times \pi_2) = L(s, \pi_1 \times \pi_2 \times \pi_3)$$

has no Landau-Siegel zero by Lemma 2.8 (2).

(b) Assume otherwise that neither $\pi_1 \boxtimes \pi_3$ nor $\pi_1 \boxtimes \pi_2$ is cuspidal. Then by Lemma 2.3 (1), we have $B_F^K(\pi_2) \cong B_F^K(\pi_2) \otimes \chi_1^\theta\chi_1^{-1}$ and $B_F^K(\pi_3) \cong B_F^K(\pi_3) \otimes \chi_1^\theta\chi_1^{-1}$. Let L/K be the quadratic extension associated with the quadratic character $\chi_1^\theta\chi_1^{-1}$

on K. Then $B_F^K(\pi_2)\otimes\chi_1 = I_L^K(\xi_2)$ and $B_F^K(\pi_3) = I_L^K(\xi_3)$ for characters ξ_2,ξ_3 on L. In this case we have

$$\begin{aligned} L(s,\pi_1\times\pi_2\times\pi_3) &= L(s, B_F^K(\pi_2)\times B_F^K(\pi_3)\otimes\chi_1)\\ &= L(s, I_L^K(\xi_2)\times I_L^K(\xi_3))\\ &= L(s,\xi_3\xi_2)\cdot L(s,\xi_3\xi_2^{\theta_{L/K}}). \end{aligned}$$

Thus $L(s,\pi_1\times\pi_2\times\pi_3)$ has no Landau-Siegel zero unless $(\xi_3\xi_2)^2=1$ or $(\xi_3\xi_2^{\theta_{L/K}})^2=1$. Moreover, there is at most one Landau-Siegel zero, by the same argument as in the proof of [RW03, Theorem A]. This proves Theorem 3.1 (2).

(3) Assume that neither $B_F^K(\pi_2)$ nor $B_F^K(\pi_3)$ is cuspidal. Then $\pi_2 = I_K^F(\chi_2)$ and $\pi_3 = I_K^F(\chi_3)$ for characters χ_2,χ_3 on F. By Lemma 2.1 we have $B_F^K(\pi_2) = \chi_2\boxplus\chi_2^\theta$ and $B_F^K(\pi_3) = \chi_3\boxplus\chi_3^\theta$. In this case we have

$$\begin{aligned} L(s,\pi_1\times\pi_2\times\pi_3) &= L(s,(\chi_2\boxplus\chi_2^\theta)\times(\chi_3\boxplus\chi_3^\theta)\otimes\chi_1)\\ &= L(s,\chi_1\chi_2\chi_3)\cdot L(s,\chi_1\chi_2\chi_3^\theta)\cdot L(s,\chi_1\chi_2^\theta\chi_3)\cdot L(s,\chi_1\chi_2^\theta\chi_3^\theta). \end{aligned}$$

A priori there are at most four Landau-Siegel zeros, if the four characters $\chi_1\chi_2\chi_3, \chi_1\chi_2\chi_3^\theta, \chi_1\chi_2^\theta\chi_3$ and $\chi_1\chi_2^\theta\chi_3^\theta$ are all real. However, we can generalize the argument in [RW03, Section 4] to show that there are at most two (and in most cases, even at most one) Landau-Siegel zeros. We discuss the following cases depending on the number of real characters among $\chi_1\chi_2\chi_3, \chi_1\chi_2\chi_3^\theta, \chi_1\chi_2^\theta\chi_3$ and $\chi_1\chi_2^\theta\chi_3^\theta$.

(a) Assume there is at most one real character. Then $L(s,\pi_1\times\pi_2\times\pi_3)$ has at most one Landau-Siegel zero, since L-functions of complex characters have no Landau-Siegel zero by Lemma 2.5 (1).

(b) Assume there are exactly two real characters, say $\chi_1\chi_2\chi_3$ and $\chi_1\chi_2\chi_3^\theta$ are real. Then one can show that $L(s,\chi_1\chi_2\chi_3)\cdot L(s,\chi_1\chi_2\chi_3^\theta)$ has at most one Landau-Siegel zero using the same argument as in [RW03, Section 4].

(c) We claim that there cannot be exactly three real characters. This is by noting that if three of them, say $\chi_1\chi_2\chi_3, \chi_1\chi_2\chi_3^\theta$ and $\chi_1\chi_2^\theta\chi_3$, are real, then the fourth character

$$\chi_1\chi_2^\theta\chi_3^\theta = \frac{\chi_1\chi_2\chi_3^\theta\cdot\chi_1\chi_2^\theta\chi_3}{\chi_1\chi_2\chi_3}$$

is also real.

(d) Assume that all four characters are real. Then we further discuss by counting the number of trivial characters.

(i) Assume further that there is at least one trivial character, say $\chi_1\chi_2\chi_3 = 1$. We set $\mu := \chi_1\chi_2\chi_3^\theta$ and $\nu := \chi_1\chi_2^\theta\chi_3$. Then

$$\chi_1\chi_2^\theta\chi_3^\theta = \frac{\chi_1\chi_2\chi_3^\theta\cdot\chi_1\chi_2^\theta\chi_3}{\chi_1\chi_2\chi_3} = \mu\nu.$$

In this case we have

$$L(s,\pi_1\times\pi_2\times\pi_3) = \zeta_K(s)L(s,\mu)L(s,\nu)L(s,\mu\nu).$$

Note that since $\chi_2\neq\chi_2^\theta$ and $\chi_3\neq\chi_3^\theta$, μ and ν are both non-trivial quadratic. Thus $L(s,\pi_1\times\pi_2\times\pi_3)$ is a Dirichlet series with non-negative coefficients, that has a pole of order 2 or 1 at $s=1$, depending on whether $\mu\nu=1$ or not (equivalently, whether $(\chi_2\chi_3)^\theta = \chi_2\chi_3$ or not).

Thus $L(s,\pi_1\times\pi_2\times\pi_3)$ has at most two Landau-Siegel zeros (or at most one Landau-Siegel zero, if $(\chi_2\chi_3)^\theta\neq\chi_2\chi_3$).

(ii) Assume otherwise that all four characters are non-trivial quadratic. We set $\alpha:=\chi_1\chi_2\chi_3$, $\beta:=\chi_3^\theta/\chi_3$ and $\gamma:=\chi_2^\theta/\chi_2$. Note that by our assumption α,β,γ are all non-trivial quadratic. We have the following relations:

$$\chi_1\chi_2\chi_3^\theta=\alpha\beta,$$
$$\chi_1\chi_2^\theta\chi_3=\alpha\gamma,$$
$$\chi_1\chi_2^\theta\chi_3^\theta=\alpha\beta\gamma.$$

In this case we have

$$L(s,\pi_1\times\pi_2\times\pi_3)=L(s,\alpha)L(s,\alpha\beta)L(s,\alpha\gamma)L(s,\alpha\beta\gamma).$$

Consider the auxiliary L-function

$$L(s):=\zeta_K(s)L(s,\alpha)L(s,\beta)L(s,\gamma)L(s,\alpha\beta)L(s,\alpha\gamma)L(s,\beta\gamma)L(s,\alpha\beta\gamma),$$

which is a Dirichlet series with non-negative coefficients, that has a pole of order 2 or 1 at $s=1$, depending on whether $\beta\gamma=1$ or not (equivalently, whether $(\chi_2\chi_3)^\theta=\chi_2\chi_3$ or not). Thus $L(s,\pi_1\times\pi_2\times\pi_3)$, as part of $L(s)$, has at most two Landau-Siegel zeros (or at most one Landau-Siegel zero, if $(\chi_2\chi_3)^\theta\neq\chi_2\chi_3$). We have thus finished the proof of Theorem 3.1 (1).

4. Proof of Theorem B

The goal of this section is to prove the following refined version of Theorem B:

Theorem 4.1. *Let $\pi_1,\pi_2\in\mathcal{A}_0(2,F)$ and $\pi\in\mathcal{A}_0(3,F)$ with central characters ω_1,ω_2 and ω respectively. Then $L(s,\pi_1\times\pi_2\times\pi)$ has no Landau-Siegel zero except possibly in the following cases:*

(1) π_1,π_2 are non-dihedral twist-equivalent, and π is twist-equivalent to $Ad(\pi_1)$. Moreover, write $\pi_2\cong\pi_1\otimes\eta$ and $\pi\otimes\omega_1\eta\cong Ad(\pi_1)\otimes\mu$ for character η and μ on F. Then there is at most one Landau-Siegel zero coming form $L(s,\mu)$, which can only occur if $\mu^2=1$.

(2) π_1,π_2 are not of solvable polyhedral type, and are not twist-equivalent, but π is twist-equivalent to either $Ad(\pi_1)$ or $Ad(\pi_2)$, say $Ad(\pi_1)$. Moreover, write $\pi\cong Ad(\pi_1)\otimes\eta$ for some character η on F. Then there is no Landau-Siegel zero if $A^3(\pi_2\otimes\eta)$ is not isomorphic to $\overline{A^3(\pi_1)}$.

Remark 4.1. *The condition in Theorem 4.1 (2) is satisfied if F is totally real, and π_1,π_2 are regular algebraic by* [Tho25b, Lemma 4.2]. *Thus there is no Landau-Siegel zero in this case.*

4.1. **General Type.** Assume (π_1,π_2,π) is of general type. Recall that this means π_1,π_2 are non-dihedral, not twist-equivalent, and π is not twist-equivalent to either $\mathrm{Ad}(\pi_1)$ or $\mathrm{Ad}(\pi_2)$. We set

$$\Pi:=(\pi_1\boxtimes\pi)\boxplus\overline{\pi_2}\boxplus(\mathrm{Ad}(\pi_1)\boxtimes\overline{\pi_2}).$$

Now since π_1 is non-dihedral and π is not twist-equivalent to $\mathrm{Ad}(\pi_1)$, by Lemma 2.3 (2), we know that $\pi_1\boxtimes\pi$ is cuspidal. Moreover, since π_1 and π_2 are not twist-equivalent, by Lemma 2.4 we know that $\mathrm{Ad}(\pi_1)$ and $\mathrm{Ad}(\overline{\pi_2})=\mathrm{Ad}(\pi_2)$ are not twist-equivalent either. Therefore

$\mathrm{Ad}(\pi_1)\boxtimes\overline{\pi_2}$ is also cuspidal. We then claim that $\pi_1\boxtimes\pi$ and $\mathrm{Ad}(\pi_1)\boxtimes\overline{\pi_2}$ are not isomorphic. To do this we compute the Rankin-Selberg L-function $L(s,(\pi_1\boxtimes\pi)\times\overline{(\mathrm{Ad}(\pi_1)\boxtimes\overline{\pi_2})})$:

$$\begin{aligned}L(s,(\pi_1\boxtimes\pi)\times\overline{(\mathrm{Ad}(\pi_1)\boxtimes\overline{\pi_2})}) &= L(s,(\pi_1\boxtimes\pi)\times(\mathrm{Ad}(\pi_1)\boxtimes\pi_2))\\ &= L(s,(\pi_1\boxtimes\mathrm{Ad}(\pi_1))\times(\pi_2\boxtimes\pi))\\ &= L(s,(\pi_1\boxplus A^3(\pi_1))\times(\pi_2\boxtimes\pi))\\ &= L(s,\pi_1\times(\pi_2\boxtimes\pi))\cdot L(s,A^3(\pi_1)\times(\pi_2\boxtimes\pi)).\end{aligned}$$

Now since π_2 is non-dihedral and π is not twist-equivalent to $\mathrm{Ad}(\pi_2)$, by Lemma 2.3 (2), $\pi_2\boxtimes\pi$ is cuspidal of degree 6, which is not isomorphic to either $\overline{\pi_1}$ or any isobaric constitute of $\overline{A^3(\pi_1)}$. Therefore both $L(s,\pi_1\times(\pi_2\boxtimes\pi))$ and $L(s,A^3(\pi_1)\times(\pi_2\boxtimes\pi))$ are entire. Thus $\pi_1\boxtimes\pi$ is not isomorphic to $\mathrm{Ad}(\pi_1)\boxtimes\overline{\pi_2}$. In view of the decomposition above we have

$$\begin{aligned}L(s,\Pi\times\overline{\Pi}) = {}& L(s,\pi_1\times\pi_2\times\pi)^2\cdot L(s,\overline{\pi_1}\times\overline{\pi_2}\times\overline{\pi})^2\\ &\cdot L(s,A^3(\pi_1)\times(\pi_2\boxtimes\pi))\cdot L(s,\overline{A^3(\pi_1)}\times(\overline{\pi_2\boxtimes\pi}))\\ &\cdot L(s,\pi_2\times(\mathrm{Ad}(\pi_1)\boxtimes\overline{\pi_2}))^2\\ &\cdot L(s,(\pi_1\boxtimes\pi)\times(\overline{\pi_1\boxtimes\pi}))\cdot L(s,\pi_2\times\overline{\pi_2})\cdot L(s,(\mathrm{Ad}(\pi_1)\boxtimes\pi_2)\times(\overline{\mathrm{Ad}(\pi_1)\boxtimes\pi_2})).\end{aligned}$$

By Lemma 2.7, $L(s,\pi_1\times\pi_2\times\pi)$ has no Landau-Siegel zero.

4.2. **Twist-equivalent Type.** Assume (π_1,π_2,π) is of twist-equivalent type. Recall that this means π_1,π_2 are both non-dihedral, but either (1) π_1 is twist-equivalent to π_2, or (2) π is twist-equivalent to $\mathrm{Ad}(\pi_1)$ or $\mathrm{Ad}(\pi_2)$. Thus we have two cases to consider:

(1) Assume that π_1 is twist-equivalent to π_2. Then $\pi_2\cong\pi_1\otimes\eta$ for some character η on F. In this case $L(s,\pi_1\times\pi_2\times\pi)$ decomposes as

$$\begin{aligned}L(s,\pi_1\times\pi_2\times\pi) &= L(s,\pi_1\times(\pi_1\otimes\eta)\times\pi)\\ &= L(s,(\pi_1\boxtimes\pi_1)\times\pi\otimes\eta)\\ &= L(s,(\mathrm{Ad}(\pi_1)\otimes\omega_1\boxplus\omega_1)\times\pi\otimes\eta)\\ &= L(s,\pi\otimes\omega_1\eta)\cdot L(s,\mathrm{Ad}(\pi_1)\times(\pi\otimes\omega_1\eta)).\end{aligned}$$

Now $L(s,\pi\otimes\omega_1\eta)$ has no Landau-Siegel zero by Lemma 2.5 (3). If $\pi\otimes\omega_1\eta$ is non-self-dual, then $L(s,\mathrm{Ad}(\pi_1)\times(\pi\otimes\omega_1\eta))$ has no Landau-Siegel zero by Lemma 2.8 (1). Therefore $L(s,\pi_1\times\pi_2\times\pi)$ has no Landau-Siegel zero in this case. If $\pi\otimes\omega_1\eta$ is self-dual, then by [Ram14, Theorem A] we have $\pi\otimes\omega_1\eta\cong\mathrm{Ad}(\pi_3)\otimes\mu$ for some non-dihedral $\pi_3\in\mathcal{A}_0(2,F)$ and some character μ on F with $\mu^2=1$. We are thus reduced to study Landau-Siegel zeros of the Rankin-Selberg L-function

$$L(s,\mathrm{Ad}(\pi_1)\times\mathrm{Ad}(\pi_3)\otimes\mu),$$

which has been studied in [Tho25a]. We will briefly record the proof here for completeness.

(a) Assume further that π_1 is not twist-equivalent to π_3. Consider the following auxiliary L-function

$$\begin{aligned}L(s) :=\ & L(s, \mathrm{Ad}(\pi_1) \times \mathrm{Ad}(\pi_3) \otimes \mu)^8 \\ & \cdot L(s, A^4(\pi_1) \times \mathrm{Ad}(\pi_3) \otimes \mu)^4 \cdot L(s, \mathrm{Ad}(\pi_3) \otimes \mu)^4 \\ & \cdot L(s, \mathrm{Ad}(\pi_1) \times A^4(\pi_3))^2 \cdot L(s, \mathrm{Ad}(\pi_1) \times \mathrm{Ad}(\pi_3))^2 \cdot L(s, \mathrm{Ad}(\pi_1))^2 \\ & \cdot L(s, \mathrm{Ad}(\pi_1) \times \mathrm{Ad}(\pi_1))^4 \cdot L(s, \mathrm{Ad}(\pi_3) \times \mathrm{Ad}(\pi_3)) \\ & \cdot L(s, (\mathrm{Ad}(\pi_1) \boxtimes \mathrm{Ad}(\pi_1)) \times (\mathrm{Ad}(\pi_3) \boxtimes \mathrm{Ad}(\pi_3))).\end{aligned}$$

Let us write

$$L(s) = \prod_v \prod_{j=1}^{d_L} \left(1 - \frac{\alpha_{L,j}(v)}{q_v^s}\right)^{-1},$$

$$-\frac{L'}{L}(s) = \sum_v \sum_{\ell=0}^{\infty} \frac{a_L(v^\ell)}{q_v^{\ell s}} \cdot \log(q_v),$$

where d_L is the degree of $L(s)$, v runs through finite places of F, and $q_v = \#F_v/\mathcal{O}_v$ is the order of the residue field at v. The following two facts can be verified for $L(s)$:

(i) For each finite place v where π_1, π_3 and μ are unramified, we have

$$a_L(v^\ell) = |2a_{\mathrm{Ad}(\pi_1)}(v^\ell) + a_{\mathrm{Ad}(\pi_3)\otimes\mu}(v^\ell) + a_{\mathrm{Ad}(\pi_1)\times\mathrm{Ad}(\pi_3)\otimes\mu}(v^\ell)|^2 \geq 0.$$

(ii) For each finite place v, we have $\alpha_{L,j}(v) \ll q_v^{\frac{8}{9}}$.

The fact (i) is a direct computation and can be found in [Tho25a, Lemma 3.2]. The fact (ii) is due to the bound $|\alpha_{\pi_1,j}(v)|, |\alpha_{\pi_3,j}(v)| \leq q_v^{\frac{1}{9}}$ [KS02a, Proposition 4.8]. Thus by [IK04, Lemma 5.9], $L(s)$ has at most r Landau-Siegel zeros, where r is the order of pole of $L(s)$ at $s = 1$. Let us proceed to count the order of pole of $L(s)$ at $s = 1$, and compare it to the multiplicity of the factor $L(s, \mathrm{Ad}(\pi_1) \times \mathrm{Ad}(\pi_3) \otimes \mu)$. To do this we first apply the decomposition

$$\mathrm{Ad}(\pi_1) \boxtimes \mathrm{Ad}(\pi_1) = A^4(\pi_1) \boxplus \mathrm{Ad}(\pi_1) \boxplus 1$$

to further write $L(s)$ as

$$\begin{aligned}L(s) :=\ & L(s, \mathrm{Ad}(\pi_1) \times \mathrm{Ad}(\pi_3) \otimes \mu)^8 \\ & \cdot L(s, A^4(\pi_1) \times \mathrm{Ad}(\pi_3) \otimes \mu)^4 \cdot L(s, \mathrm{Ad}(\pi_3) \otimes \mu)^4 \\ & \cdot L(s, \mathrm{Ad}(\pi_1) \times A^4(\pi_3))^3 \cdot L(s, \mathrm{Ad}(\pi_1) \times \mathrm{Ad}(\pi_3))^3 \cdot L(s, \mathrm{Ad}(\pi_1))^3 \\ & \cdot L(s, A^4(\pi_1) \times A^4(\pi_3)) \cdot L(s, A^4(\pi_1) \times \mathrm{Ad}(\pi_3)) \cdot L(s, A^4(\pi_1)) \\ & \cdot L(s, A^4(\pi_3)) \cdot L(s, \mathrm{Ad}(\pi_3)) \\ & \cdot L(s, \mathrm{Ad}(\pi_1) \times \mathrm{Ad}(\pi_1))^4 \cdot L(s, \mathrm{Ad}(\pi_3) \times \mathrm{Ad}(\pi_3)) \cdot \zeta_F(s).\end{aligned}$$

Note that the last line of the expression above contributes to $L(s)$ a pole at $s = 1$ of order 6. The only other factors that could possibly contribute to $L(s)$ a pole at $s = 1$ are $L(s, A^4(\pi_1) \times A^4(\pi_3))$, $L(s, \mathrm{Ad}(\pi_1) \times A^4(\pi_3))^3$, $L(s, A^4(\pi_1) \times \mathrm{Ad}(\pi_3))$ and $L(s, A^4(\pi_1) \times \mathrm{Ad}(\pi_3) \otimes \mu)^4$. However, in view of Lemma 2.2 (3), Lemma 2.4 and the assumption that π_1 is not twist-equivalent to π_3, $L(s, \mathrm{Ad}(\pi_1) \times$

$A^4(\pi_3))^3, L(s, A^4(\pi_1) \times \mathrm{Ad}(\pi_3))$ and $L(s, A^4(\pi_1) \times \mathrm{Ad}(\pi_3) \otimes \mu)^4$ are entire. Thus the only factor that my contribute to $L(s)$ additional poles at $s = 1$ is $L(s, A^4(\pi_1) \times A^4(\pi_3))$.

(i) Assume that π_1 is not of solvable polyhedral type. Then $A^4(\pi_1)$ is cuspidal. In this case the order of pole of $L(s)$ at $s = 1$ is 7 or 6, depending on whether $A^4(\pi_1) \cong A^4(\pi_3)$ or not. In either case $L(s, \mathrm{Ad}(\pi_1) \times \mathrm{Ad}(\pi_3) \otimes \mu)$ has no Landau-Siegel zero, since any Landau-Siegel zero of it would be a Landau-Siegel zero of order at least 8 of $L(s)$.

(ii) Assume that π_1 is of octahedral type. Then by Lemma 2.2 (3), $A^4(\pi_1)$ decomposes as

$$A^4(\pi_1) = (Ad(\pi_1) \otimes \nu) \boxplus I_K^F(\chi^{-1})$$

for some quadratic character ν on F and some character χ on a quadratic extension K/F. In this case $L(s, A^4(\pi_1) \times A^4(\pi_3))$ has at most 1 simple pole at $s = 1$, which only occurs if π_3 is also octahedral. Thus $r \leq 7 < 8$ and $L(s, \mathrm{Ad}(\pi_1) \times \mathrm{Ad}(\pi_3) \otimes \mu)$ has no Landau-Siegel zero.

(iii) Assume that π_1 is of tetrahedral type. Then by Lemma 2.2 (3), $A^4(\pi_1)$ decomposes as

$$A^4(\pi_1) = \mathrm{Ad}(\pi_1) \boxplus \nu \boxplus \nu^2$$

for some cubic character ν on F. In this case $L(s, A^4(\pi_1) \times A^4(\pi_3))$ can only have a pole at $s = 1$ if π_3 is also tetrahedral, and the order of the pole is at most 4. We then make use of the decomposition

$$\begin{aligned} L(s, A^4(\pi_1) \times \mathrm{Ad}(\pi_3) \otimes \mu) &= L(s, (\mathrm{Ad}(\pi_1) \boxplus \nu \boxplus \nu^2) \times \mathrm{Ad}(\pi_3) \otimes \mu) \\ &= L(s, \mathrm{Ad}(\pi_3) \otimes \mu\nu) \cdot L(s, \mathrm{Ad}(\pi_3) \otimes \mu\nu^2) \\ &\quad \cdot L(s, \mathrm{Ad}(\pi_1) \times \mathrm{Ad}(\pi_3) \otimes \mu) \end{aligned}$$

to see that any Landau-Siegel zero of $L(s, \mathrm{Ad}(\pi_1) \times \mathrm{Ad}(\pi_3) \otimes \mu)$ would be a Landau-Siegel zero of $L(s)$ of order at least 12. Since $r \leq 10 < 12$, $L(s, \mathrm{Ad}(\pi_1) \times \mathrm{Ad}(\pi_3) \otimes \mu)$ has no Landau-Siegel zero.

(b) Assume otherwise that π_1 is twist-equivalent to π_3. In this case $L(s, \mathrm{Ad}(\pi_1) \times \mathrm{Ad}(\pi_3) \otimes \mu)$ decomposes as

$$\begin{aligned} L(s, \mathrm{Ad}(\pi_1) \times \mathrm{Ad}(\pi_3) \otimes \mu) &= L(s, \mathrm{Ad}(\pi_1) \boxtimes \mathrm{Ad}(\pi_1) \otimes \mu) \\ &= L(s, (A^4(\pi_1) \boxplus \mathrm{Ad}(\pi_1) \boxplus 1) \otimes \mu) \\ &= L(s, A^4(\pi_1) \otimes \mu) \cdot L(s, \mathrm{Ad}(\pi_1) \otimes \mu) \cdot L(s, \mu). \end{aligned}$$

Now $L(s, \mathrm{Ad}(\pi_1) \otimes \mu)$ has no Landau-Siegel zero by Lemma 2.5 (3). We then prove that $L(s, A^4(\pi_1) \otimes \mu)$ has no Landau-Siegel zero.

(i) Assume that π_1 is of tetrahedral type. Then by Lemma 2.2 (3), $L(s, A^4(\pi_1) \otimes \mu)$ decomposes as

$$\begin{aligned} L(s, A^4(\pi_1) \otimes \mu) &= L(s, (\mathrm{Ad}(\pi_1) \boxplus \nu \boxplus \nu^2) \otimes \mu) \\ &= L(s, \mathrm{Ad}(\pi_1) \otimes \mu) \cdot L(s, \mu\nu) \cdot L(s, \mu\nu^2) \end{aligned}$$

for some cubic character ν on F. Now $L(s, \mathrm{Ad}(\pi_1) \otimes \mu)$ has no Landau-Siegel zero by Lemma 2.5 (3). Moreover, $L(s, \mu\nu)$ and $L(s, \mu\nu^2)$ have no

Landau-Siegel zero since $(\mu\nu)^2 = \nu^2 \neq 1$ and $(\mu\nu^2)^2 = \nu \neq 1$. Therefore $L(s, A^4(\pi_1) \otimes \mu)$ has no Landau-Siegel zero.

(ii) Assume that π_1 is of octahedral type. Then by Lemma 2.2 (3), $L(s, A^4(\pi_1) \otimes \mu)$ decomposes as

$$\begin{aligned} L(s, A^4(\pi_1) \otimes \mu) &= L(s, ((\mathrm{Ad}(\pi_1) \otimes \nu) \boxplus I_K^F(\chi^{-1})) \otimes \mu) \\ &= L(s, \mathrm{Ad}(\pi_1) \otimes \mu\nu) \cdot L(s, I_K^F(\chi^{-1}) \otimes \mu) \end{aligned}$$

for some quadratic character ν on F and some character χ on a quadratic extension K/F, which has no Landau-Siegel zero by Lemma 2.5 (3).

(iii) Assume that π_1 is not of solvable polyhedral type. Then $A^4(\pi_1)$ is cuspidal. We set

$$\Pi := 1 \boxplus (A^4(\pi_1) \otimes \mu) \boxplus \mathrm{Ad}(\pi_1)$$

and compute

$$\begin{aligned} L(s, \Pi \times \overline{\Pi}) = L(s, A^4(\pi_1) \otimes \mu)^2 \\ \cdot L(s, (A^4(\pi_1) \otimes \mu) \times \mathrm{Ad}(\pi_1))^2 \cdot L(s, \mathrm{Ad}(\pi_1))^2 \\ \cdot \zeta_F(s) \cdot L(s, \mathrm{Ad}(\pi_1) \times \mathrm{Ad}(\pi_1)) \cdot L(s, A^4(\pi_1) \times A^4(\pi_1)) \end{aligned}$$

Note that $L(s, (A^4(\pi_1) \otimes \mu) \times \mathrm{Ad}(\pi_1))$ decomposes as

$$\begin{aligned} L(s, (A^4(\pi_1) \otimes \mu) \times \mathrm{Ad}(\pi_1)) &= L(s, A^4(\pi_1) \otimes \mu) \\ &\quad \cdot L(s, \pi_1; \mathrm{Sym}^6 \otimes \omega_1^{-3}\mu) \cdot L(s, \mathrm{Ad}(\pi_1) \otimes \mu) \\ &= L(s, A^4(\pi_1) \otimes \mu) \cdot L(s, \mathrm{Sym}^3(\pi_1); \mathrm{Sym}^2 \otimes \omega_1^{-3}\mu) \end{aligned}$$

Thus $L(s, \Pi \times \overline{\Pi})$ can be further written as

$$\begin{aligned} L(s, \Pi \times \overline{\Pi}) = L(s, A^4(\pi_1) \otimes \mu)^4 \\ \cdot L(s, \mathrm{Sym}^3(\pi_1); \mathrm{Sym}^2 \otimes \omega_1^{-3}\mu)^2 \cdot L(s, \mathrm{Ad}(\pi_1))^2 \\ \cdot \zeta_F(s) \cdot L(s, \mathrm{Ad}(\pi_1) \times \mathrm{Ad}(\pi_1)) \cdot L(s, A^4(\pi_1) \times A^4(\pi_1)). \end{aligned}$$

Let S be the set of finite places of F where π_1 or μ is ramified. By [Tak14, Corollary 5.8], the partial L-function $L^S(s, \mathrm{Sym}^3(\pi_1); \mathrm{Sym}^2 \otimes \omega_1^{-3}\mu)$ is holomorphic in $(7/8, 1)$. Moreover, the finite product of local L-functions

$$\prod_{v \in S} L(s, \mathrm{Sym}^3(\pi_{1,v}); \mathrm{Sym}^2 \otimes \omega_{1,v}1^{-3}\mu_v) = \prod_{v \in S} \frac{L(s, A^4(\pi_{1,v}) \times \mathrm{Ad}(\pi_{1,v}) \otimes \mu_v)}{L(s, A^4(\pi_{1,v}) \otimes \mu_v)}$$

is holomorphic in $(56/65, 1)$ by [LRS99, Theorem 2]. Thus by Lemma 2.7, $L(s, A^4(\pi_1) \otimes \mu)$ has no Landau-Siegel zero.

We have thus eliminated Landau-Siegel zeros of $L(s, A^4(\pi_1) \otimes \mu)$ in all cases. Therefore the only possible Landau-Siegel zero of $L(s, \mathrm{Ad}(\pi_1) \times \mathrm{Ad}(\pi_3) \otimes \mu)$ comes from $L(s, \mu)$. This proves Theorem 4.1 (1).

(2) Assume π_1, π_2 are not twist-equivalent, and that π is twist-equivalent to $\mathrm{Ad}(\pi_1)$ or $\mathrm{Ad}(\pi_2)$, say $\pi \cong \mathrm{Ad}(\pi_1) \otimes \eta$ for some character η on F. In this case $L(s, \pi_1 \times \pi_2 \times \pi)$

decomposes as

$$\begin{aligned} L(s,\pi_1\times\pi_2\times\pi) &= L(s,\pi_1\times\pi_2\times(\mathrm{Ad}(\pi_1)\otimes\eta))\\ &= L(s,(\pi_1\boxtimes\mathrm{Ad}(\pi_1))\times\pi_2\otimes\eta)\\ &= L(s,(\pi_1\boxplus A^3(\pi_1))\times\pi_2\otimes\eta)\\ &= L(s,\pi_1\times\pi_2\otimes\eta)\cdot L(s,A^3(\pi_1)\times\pi_2\otimes\eta). \end{aligned}$$

Now since π_1,π_2 are non-dihedral and are not twist-equivalent, $L(s,\pi_1\times\pi_2\otimes\eta)$ has no Landau-Siegel zero by Lemma 2.6 (1). We then discuss Landau-Siegel zeros of $L(s,A^3(\pi_1)\times\pi_2\otimes\eta)$:

(a) Assume π_1 is tetrahedral. Then by Lemma 2.2 (2), $A^3(\pi_1)$ decomposes as

$$A^3(\pi_1)=(\pi_1\otimes\nu)\boxplus(\pi_1\otimes\nu^2)$$

for some cubic character ν on F. Thus we have

$$L(s,A^3(\pi_1)\times\pi_2\otimes\eta)=L(s,\pi_1\times\pi_2\otimes\eta\nu)\cdot L(s,\pi_1\times\pi_2\otimes\eta\nu^2)$$

which has no Landau-Siegel by Lemma 2.6 (1).

(b) Assume π_1 is octahedral, i.e., $A^3(\pi_1)$ is cuspidal and monomial. Write $A^3(\pi_1)\cong A^3(\pi_1)\otimes\mu$ for some quadratic character μ on F. Since π_2 is non-dihedral, $\pi_2\otimes\eta$ is not isomorphic to $(\pi_2\otimes\eta)\otimes\mu$. Therefore $L(s,A^3(\pi_1)\times\pi_2\otimes\eta)$ has no Landau-Siegel zero by Lemma 2.8 (2).

(c) Assume π_1 is not of solvable polyhedral type. In this case we go back to the triple product L-function

$$L(s,\pi_1\times\pi_2\times\pi)=L(s,\pi_1\times\pi_2'\times\mathrm{Ad}(\pi_1)).$$

Here we write $\pi_2'=\pi_2\otimes\eta$. We set

$$\Pi:=(\pi_2'\boxtimes\mathrm{Ad}(\pi_1))\boxplus\overline{\pi_1}\boxplus(\overline{\pi_1}\boxtimes\mathrm{Ad}(\pi_2')).$$

To test whether $\pi_2'\boxtimes\mathrm{Ad}(\pi_1)$ and $\overline{\pi_1}\boxtimes\mathrm{Ad}(\pi_2')$ are isomorphic we compute

$$\begin{aligned} L(s,(\pi_2'\boxtimes\mathrm{Ad}(\pi_1))\times(\pi_1\boxtimes\mathrm{Ad}(\pi_2'))) &= L(s,(\pi_1\boxtimes\mathrm{Ad}(\pi_1))\times(\pi_2'\boxtimes\mathrm{Ad}(\pi_2')))\\ &= L(s,(\pi_1\boxplus A^3(\pi_1))\times(\pi_2'\boxplus A^3(\pi_2')))\\ &= L(s,\pi_1\times\pi_2')\cdot L(s,A^3(\pi_1)\times A^3(\pi_2'))\\ &\quad\cdot L(s,\pi_1\times A^3(\pi_2'))\cdot L(s,\pi_2'\times A^3(\pi_1)), \end{aligned}$$

which can only have a simple pole at $s=1$ if $A^3(\pi_2')\cong\overline{A^3(\pi_1)}$, which would imply that π_2 is not of solvable polyhedral type. By the decomposition above we have

$$\begin{aligned} L(s,\Pi\times\overline{\Pi}) &= L(s,\pi_1\times\pi_2'\times\mathrm{Ad}(\pi_1))^2\cdot L(s,\overline{\pi_1}\times\overline{\pi_2'}\times\mathrm{Ad}(\pi_1))^2\\ &\quad\cdot L(s,\pi_1\times A^3(\pi_2'))\cdot L(s,\overline{\pi_1}\times\overline{A^3(\pi_2')})\\ &\quad\cdot L(s,A^3(\pi_1)\times A^3(\pi_2'))\cdot L(s,\overline{A^3(\pi_1)}\times\overline{A^3(\pi_2')})\\ &\quad\cdot L(s,\pi_1\times(\overline{\pi_1}\boxtimes\mathrm{Ad}(\pi_2')))^2\\ &\quad\cdot L(s,(\pi_2'\boxtimes\mathrm{Ad}(\pi_1))\times(\overline{\pi_2'}\boxtimes\mathrm{Ad}(\pi_1)))\cdot L(s,\pi_1\times\overline{\pi_1})\\ &\quad\cdot L(s,(\pi_1\boxtimes\mathrm{Ad}(\pi_2'))\times(\overline{\pi_1}\boxtimes\mathrm{Ad}(\pi_2'))). \end{aligned}$$

Thus by Lemma 2.7, $L(s, \pi_1 \times \pi_2 \times \pi)$ can only have Landau-Siegel zeros if π_2 is not of solvable polyhedral type, and that

$$A^3(\pi_2 \otimes \eta) \cong \overline{A^3(\pi_1)}.$$

This proves Theorem 4.1 (2).

4.3. **Dihedral Type.** Assume (π_1, π_2, π) is of dihedral type. Recall that this means π_1 or π_2, say π_1, is dihedral. Write $\pi_1 = I_K^F(\chi_1)$ for a character χ_1 on some quadratic extension K/F. Then we have

$$\begin{aligned} L(s, \pi_1 \times \pi_2 \times \pi) &= L(s, (I_K^F(\chi_1) \boxtimes \pi_2) \times \pi) \\ &= L(s, I_K^F(B_F^K(\pi_2) \otimes \chi_1) \times \pi) \\ &= L(s, B_F^K(\pi_2) \times B_F^K(\pi) \otimes \chi_1). \end{aligned}$$

Let $\eta = \eta_{K/F}$ be the quadratic character on F associated with K/F. Let $\theta = \theta_{K/F}$ be the non-trivial element in $\mathrm{Gal}(K/F)$. Then π is not isomorphic to $\pi \otimes \eta$. By Lemma 2.1 we know that $B_F^K(\pi)$ is cuspidal.

(1) Assume that $B_F^K(\pi_2)$ is not cuspidal. Then $\pi_2 = I_K^F(\chi_2)$ for some character χ_2 on F. In this case we have

$$\begin{aligned} L(s, \pi_1 \times \pi_2 \times \pi) &= L(s, (\chi_2 \boxplus \chi_2^\theta) \times B_F^K(\pi) \otimes \chi_1) \\ &= L(s, B_F^K(\pi) \otimes \chi_1\chi_2) \cdot L(s, B_F^K(\pi) \otimes \chi_1\chi_2^\theta) \end{aligned}$$

which has no Landau-Siegel zero by Lemma 2.5 (3).

(2) Assume that $B_F^K(\pi_2)$ is cuspidal, but is not isomorphic to $B_F^K(\pi_2) \otimes \chi_1^\theta\chi_1^{-1}$. Then by Lemma 2.3 (1) we know that $\pi_1 \boxtimes \pi_2$ is cuspidal. We write

$$L(s, \pi_1 \times \pi_2 \times \pi) = L(s, (\pi_1 \boxtimes \pi_2) \times \pi).$$

Note that $(\pi_1 \boxtimes \pi_2) \otimes \eta = (\pi_1 \otimes \eta) \boxtimes \pi_2 = \pi_1 \boxtimes \pi_2$, and π is not isomorphic to $\pi \otimes \eta$. By Lemma 2.8 (2), $L(s, (\pi_1 \boxtimes \pi_2) \times \pi)$ has no Landau-Siegel zero.

(3) Assume that $B_F^K(\pi_2)$ is cuspidal, and is isomorphic to $B_F^K(\pi_2) \otimes \chi_1^\theta\chi_1^{-1}$. Let L/K be the quadratic extension associated with $\chi_1^\theta\chi_1^{-1}$. Then $B_F^K(\pi_2) = I_L^K(\xi_2)$ for some character ξ_2 on L. In this case we have

$$\begin{aligned} L(s, \pi_1 \times \pi_2 \times \pi) &= L(s, I_L^K(\xi_2) \times B_F^K(\pi) \otimes \chi_1) \\ &= L(s, B_K^L(B_F^K(\pi) \otimes \chi_1) \otimes \xi_2) \end{aligned}$$

which has no Landau-Siegel zero by Lemma 2.5 (3).

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